

Cse:475

Section: 6

Group members:

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Group no: 10

Dataset no:5

Dataset Name:

Bot-IoT

Dataset: <https://research.unsw.edu.au/projects/bot-iot-dataset>

Best method: GNN (flow-as-edge)

Authors(Year)	Title	Dataset	Application Area	Method/ model	Metrics	Strengths	Limitation/ gap	Relation to our work
1. Wai Weng Lo, Siamak Layeghy, Mohanad Sarhan, Marcus Gallagher, and Marius Portmann (2022)	E-GraphSAGE : A Graph Neural Network based Intrusion Detection System for IoT	BoT-IoT, ToN-IoT, NF-BoT-IoT, NF-ToN-IoT	Network Intrusion Detection Systems (NIDS) for Internet of Things (IoT)	E-GraphSAGE (An extended GraphSAGE architecture that incorporates edge flow features alongside topological information for	Accuracy, Precision, F1-Score, Detection Rate (Recall), False Alarm Rate (FAR), Weighted F1-Score	Leverages both topological graph structures and edge features directly for edge classification. Uses an inductive learning strategy, enabling it to classify	High computational complexity with full neighborhood aggregation on large-scale graphs. Does not currently utilize non-uniform or explainable	Demonstrates the effectiveness of GNN edge-embedding techniques over traditional flow-based network traffic independent models, serving as a primary benchmark

				edge classification via inductive representation learning)		unseen host nodes/IPs during deployment. Outperforms state-of-the-art methods across four comprehensive IoT benchmark datasets.	graph sampling models.	for GraphSAGE extensions in IoT intrusion detection.
2. Tien Ngo, Jiao Yin, Yong-Feng Ge, and Hua Wang (2025)	Optimizing IoT Intrusion Detection—A Graph Neural Network Approach with Attribute-Based Graph Construction	NF-BoT -IoT, NF-ToN -IoT	IoT Network Intrusion Detection Systems (NIDS) / Cybersecurity	TKSGF (Top-K Similarity Graph Framework using Cosine Similarity and FAISS) combined with GraphSAGE , GCN , and GAT architectures	Average F1-Score, Max F1-Score, Standard Deviation, Average Run Time	Replaces physical connection topology with attribute-based similarity graphs, resulting in more semantically meaningful node connections. Accelerates k-NN graph construction using GPU-optimized FAISS vector search. Provides extensive empirical analysis across graph directionality (directed vs.	GAT model performance severely degrades in scale and computation speed when handling large dynamic graphs. Heavy dependency on initial hyperparameter selection (such as similarity thresholds and top-\$k\$ bounds)	Demonstrates that similarity-based topological graphs outperform direct physical link networks for learning flow representations, providing insights on hyperparameter selection and framework scaling for GNN-based IoT security systems.

						undirected), and GNN backbones.		
3. Yalu Wang, Zhijie Han, Jie Li, Xin He (2023)	BS-GAT: Behavior Similarity Based Graph Attention Network for Network Intrusion Detection	NF-BoT -IoT-v2 and NF-ToN -IoT-v2	IoT Network Intrusion Detection / Edge Security	BS-GAT (Behavior Similarity Graph Attention Network) — uses a 4-step hierarchical behavior matching rule (IPs, subnets, ports, prefixes) to construct edges and incorporates explicit behavioral weights into GAT self-attention.	Accuracy, Precision, Recall, F1-Score, and Gini Coefficient (neighbor distribution balance).	Prevents feature blurring and overfitting caused by node degree imbalance (achieves low Gini coefficients of \$0.3529\$ and \$0.4203\$). Maintains high multi-class detection stability across minority attack classes.	High graph construction time complexity on large-scale flow datasets.	Demonstrates how domain-specific behavior similarity rules can prevent degree imbalance in flow graphs, serving as a benchmark for custom edge-weighting strategies.
4. Wasswa et al. (2025/2026)	Graph Attention Neural Network for Botnet Detection: Evaluating Autoencoder, VAE and PCA-Based Dimensionality Reduction	CICIoT2022 and N-BalIoT	IoT Botnet Detection & Resource-Constrained Intrusion Detection	Combines a Variational Autoencoder (VAE) for latent space feature reduction (e.g., from 84/115 features down to 8 dimensions) followed by k-NN	Accuracy, Precision, Recall, F1-Score, Inference Latency, Model Storage Size.	Significantly reduces computational complexity and memory footprint by compressing high-dimensional tabular features prior to graph construction.	Latent feature compression leads to notable drops in recall for minority attack classes (e.g., brute force and flooding attacks).	Illustrates the computational trade-offs of latent feature compression prior to graph construction, underscoring the necessity of preserving fine-grained features for rare attack types.

	sion Reduct ion			graph generatio n and GAT classificat ion.				
5. Riko Luša, Damir Pintar, and Mihaela Vranić	TE-G- SAGE: Explain able Edge-A ware Graph Neural Networ ks for Networ k Intrusio n Detecti on	NF-UN SW-NB 15-v3	Network Intrusion Detection Systems (NIDS) / Cybersecurit y Anomaly & Threat Detection.	An inductive, edge-awa re GraphSA GE-based model designed for edge classificat ion on temporal flow graphs. Preserves strict chronolog ical evaluatio n (60% train / 30% val / 10% test chronolog ical splits) to eliminate data leakage.	Recall Precisio n, Recall- AUC / PR-AU C (suited for heavy class imbalan ce) F1-Scor e Feature Attributi on & Interpret ability Scores via SHAP values	Temporal Integrity: Enforces strictly chronologi cal splits and temporal graph modeling, preventing future-dat a leakage common in random cross-vali dation. Relational Awarenes s: Leverages structural multi-hop connectivit y (2-hop neighborh oods) alongside flow-level	Evaluation Scope: Primary empirical validation was conducted on a single main benchmark dataset (NF-UNSW- NB15-v3). Computatio nal Overhead: Generating multi-hop computatio nal subgraphs and SHAP explanatio ns for every prediction adds processing latency in high-throug	TE-G-SAGE explicitly highlights the common pitfall of random train/test splits in flow-based NIDS, which introduce temporal data leakage. By adopting a time-respecti ng evaluation setup (chronologic al 60/30/10 split) on NetFlow traffic datasets like NF-UNSW-N B15-v3, TE-G-SAGE sets a

				Uses a adapted variable-level SHAP (SHapley Additive exPlanations) framework over two-hop computational subgraphs to explain predictions globally and locally.		statistical features. High Interpretability: Integrates SHAP directly into the edge-aware GNN computational graph, providing explicit feature attributions to support security analysts during incident triage. Inductive Capability: Generalizes to unseen network entities without requiring full graph retraining.	hput real-time environments. Static Snapshot Windows: Relies on structured temporal splits/batches rather than continuous real-time graph stream processing.	rigorous precedent for realistic deployment testing. Our work follows this methodology to ensure temporally valid performance metrics.
6. Ruisheng Li, Huimin Shen (Corresponding Author), and Qilong Zhang	A Dual-Perspective Self-Supervised IoT Intrusion Detection Method	ToN-IoT, NF-BoT-IoT, NF-CSE-CIC-ID S2018-v2, and NF-UNSW-NB15-v2	Network Intrusion Detection Systems (NIDS) / Cybersecurity in Internet of Things (IoT) environments	• Data Augmentation: Dual-perspective graph loss generation (Topology	F1	• Self-Supervised Representation: Learns rich structural and edge-level	Centralized Assumption : Evaluated primarily under centralized training, which may face latency and	Provides a strong reference architecture for lightweight, graph-based self-supervised learning on flow data, demonstrating how graph

	Based on Topology Reconstruction and Feature Perturbation			Reconstruction via edge sequence shuffling + Feature Perturbation via edge attribute operations like square/square root) • Graph Encoder: EE-GraphSAGE (Edge-Enhanced GraphSAGE for joint node-edge aggregation) • Loss Function: Adaptive Cosine Loss (ACos Loss) combining BCE with direction-aware cosine		representations without requiring massive labeled datasets. • Effective Edge Preservation: Leverages flow-level metadata in edge features better than traditional GNNs that discard edge information. • Lightweight & Fast: Low memory footprint and fast inference suitable for resource-constrained edge/IoT devices.	bandwidth constraints in real-world distributed edge topologies. • Static Snapshot Dependency: Graph construction relies on time-window flow aggregation, which might fail to capture fine-grained temporal dynamic shifts. • Domain Portability: Highly customized augmentation functions (like 20% feature perturbation) may require manual retuning for networks with vastly different protocol	contrastive/augmentation techniques can replace expensive supervised labeling while keeping models compact enough for real-time edge deployment
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				similarity Classifier: LightGBM downstream model		Noise Robustness: Adaptive Cosine Loss mitigates label noise and balances confidence vs. uncertain predictions.	characteristics.	
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Table: Related work papers